

Skills Summary

Choosing a Model from the Evidence

Use this reference while completing your worksheet or when studying.

Skill 1 — Spotting a linear model

Evidence: equal steps in the input give *equal* steps in the output — the **first differences** are constant.

Rule: $y = (\text{start value}) + (\text{constant step}) \times x$.

In words: a fixed amount to begin with, then the same amount added again and again.

Example: A club charges a joining fee and then a fixed amount per badge: totals are 15, 19, 23, 27 dollars for $n = 0, 1, 2, 3$ badges. Find the total for 9 badges.

Step 1. Want a rule; the first differences are 4, 4, 4 — constant — so the family is linear and I only need the start value and the step.

Step 2. Start value 15 at $n = 0$, step 4, so the rule is $15 + 4n$; at $n = 9$ this is $15 + 36$.
 $\Rightarrow$ 51 dollars

Try it: Totals are 20, 26, 32, 38 dollars for $n = 0, 1, 2, 3$. Find the total for 7.

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Skill 2 — Spotting a quadratic model

Evidence: the first differences change, but by the *same* amount each time — the **second differences** are constant. Falling-object heights and any area built from two lengths that both move behave this way.

Rule: $y = ax^2 + bx + c$; for a height in feet, $h(t) = -16t^2 + v_0t + h_0$.

Example: A ball's height in feet is 0, 40, 48, 24 at $t = 0, 1, 2, 3$ seconds, matching $h(t) = -16t^2 + 56t$. Find the height at $t = 2.5$ seconds.

Step 1. First differences 40, 8, -24 are not constant but the second differences are both -32 , so the family is quadratic and I may substitute into the quadratic rule.

Step 2. $h(2.5) = -16(6.25) + 56(2.5) = -100 + 140$.
 $\Rightarrow$ 40 feet

Try it: A second ball follows $h(t) = -16t^2 + 48t$. Find its height at $t = 2.5$ seconds.

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Skill 3 — Spotting a piecewise model

Evidence: the rule itself *changes* at a break point — a flat charge that becomes a per-unit charge, a free allowance that runs out, a rate that switches.

Rule: one expression before the break, another after it. Past a break at $x = k$:
 (amount used up to k) + (new rate) $\times$ ($x - k$).

Example: A hall costs \$5 for the first 3 hours and \$2 for every hour after that. Find the cost of 7 hours.

Step 1. The rate changes at the third hour, so the family is piecewise; 7 is past the break, so I use the second piece and count only the hours beyond 3.

Step 2. Hours past the break: $7 - 3 = 4$, so the cost is $5 + 2(4) = 5 + 8$.

$\Rightarrow$ **13** dollars

Try it: A locker costs \$4 for the first 2 days and \$3 for every day after. Find the cost of 6 days.

Check: 19

(!) Watch out: “the numbers keep going up” does not make a table linear. Rising values only rule out a decreasing rule — always test the *differences* before naming the family.